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Logistics & Distribution Study — OWL Lidl Network

Mathematical Model Formulation

TDHVRPTW — Time-Dependent Heterogeneous Vehicle Routing Problem with Time Windows

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Introduction

This report describes the mathematical model built to solve the daily delivery problem between the Dr. Oetker depot in Bielefeld and 25 Lidl stores in the OWL region. The model is classified as a Time-Dependent Heterogeneous Vehicle Routing Problem with Time Windows (TDHVRPTW). This formulation was chosen because the problem has three features that simpler models do not handle: trucks of different sizes, strict delivery time slots for each store, and travel times that change depending on the time of day due to traffic.

The network comprises 26 locations in total. Node 0 is the depot at Lutterstraße 14, Bielefeld. Nodes 1 to 25 are the Lidl stores to be supplied. The distances and travel times between all locations were derived from real road data collected through the Google Maps Distance Matrix API, giving a complete 26 × 26 matrix.

Three demand situations are tested in this project. The first is a normal working day with 99 pallets to deliver. The second is a busier day with 150 pallets. The third is a stress test with 174 pallets, designed to identify the limits of the current fleet. The mathematical structure is identical across all three situations — only the demand values change.

1. Model Parameters

Before writing the equations, the key values and rules that the model works with are defined below. These are derived directly from the operational data collected.

The Network

The model contains 26 nodes in total. Node 0 is the Dr. Oetker depot in Bielefeld, where all trucks start and finish their day. Nodes 1 through 25 are the Lidl stores spread across the OWL region. The full set of nodes is written as $V = \{0, 1, 2, \dots, 25\}$. When referring to customer nodes only, without the depot, the set $S = \{1, 2, \dots, 25\}$ is used.

Distances and Travel Times

For every pair of locations i and j, two values are defined: c_{ij}, the road distance in kilometres between them, and t_{ij}, the baseline free-flow driving time in minutes. Both were collected from the Google Maps API, so they reflect real roads rather than straight-line estimates.

Store Demands

Each Lidl store has a demand value q_i, measured in Euro pallets. In the baseline scenario S-99, stores are grouped into three demand levels: low (3 pallets), medium (4 pallets), and high (5 pallets). Across all 25 stores, the total in S-99 is 99 pallets. The depot has no demand, so $q_0 = 0$.

Service Time at Each Store

When a truck arrives at a store, it needs time to unload and complete the paperwork. This time is calculated using a formula that depends on the number of pallets being delivered:

$$s_i = 10 + 2 \times q_i \quad \text{(minutes)}$$

The 10-minute term is a fixed base covering parking, sign-in, and administrative processing. The 2-minute-per-pallet term covers physical unloading. A store receiving 4 pallets therefore requires 18 minutes of service time, and a store receiving 5 pallets requires 20 minutes.

Delivery Time Window

All stores must receive their delivery between 08:00 and 12:00. In the model, time is expressed in minutes past midnight, so the window is [480, 720]. If a truck arrives before 08:00, it waits until the store opens; if a truck cannot reach a store before 12:00, that route is infeasible.

Maximum Route Length

No truck is permitted to be out for more than 405 minutes. This is calculated from a standard 480-minute morning shift, minus a mandatory 45-minute rest break, minus a 30-minute buffer for unexpected delays. All truck types follow this same limit.

Waiting Slack

Because trucks leave at 07:00 and stores only open at 08:00, trucks reaching their first stop early must wait. A scenario-specific waiting allowance handles this: 45 minutes for S-99, 75 minutes for S-152, and 90 minutes for S-170. This does not change the delivery window itself — it allows the solver to accept routes where a truck arrives early and waits, rather than rejecting them outright.

2. The Fleet

Two types of trucks are available, together forming the heterogeneous fleet K, split into two groups:

$$K = K\_heavy \cup K\_medium$$

Truck Type	Capacity Q^k	Number Available	Used For
Heavy trucks (K_heavy)	33 pallets	4 trucks	Large delivery clusters, high-demand routes
Medium trucks (K_medium)	12 pallets	4 trucks	Urban areas, small-demand stores, short loops

This gives a total fleet capacity of 180 pallets across all 8 trucks. Each truck has its own capacity limit Q_k, which depends on which group it belongs to: a heavy truck can carry up to 33 pallets per route, and a medium truck can carry up to 12.

3. Traffic Model

One of the key features of the model is that travel times are not fixed — they depend on the time of day the truck is driving. A step function μ(t) multiplies the baseline travel times based on the elapsed time since the 07:00 departure.

$$\text{Actual travel time } (i \to j, t) = t_{ij} \times \mu(t)$$

The function μ(t) takes three values depending on the time:

Time of Day	Minutes from 07:00	μ(t)	What It Means
07:00 – 09:00	0 to 120	**1.3**	Roads are 30% slower due to morning rush hour on the A2 and B61
09:00 – 11:30	120 to 270	**1.0**	Normal speed, no significant congestion
11:30 – 12:00	270 to 300	**1.1**	A small slowdown as lunch traffic starts to build

This means all trucks departing at 07:00 face heavier traffic on their first stops; once the clock passes 09:00, routes run at normal speed. This factor was included because assuming constant travel times would produce schedules that look feasible on paper but would regularly arrive late in reality.

3a. Demand Scenarios

The model is tested under three different demand levels to understand how the system behaves as pressure increases:

Scenario	What It Represents	Total Pallets	Result
S-99	A normal weekday delivery run	99 pallets	**✓ Works**
S-152	A busy day, like before a holiday	150 pallets	**✓ Works**
S-170	A stress test at near-maximum load	174 pallets	**✗ Not feasible**

The mathematical model is identical for all three scenarios — only q_i, the demand at each store, changes. S-170 was designed deliberately to push the system past its limits. The result of that test is not a failure of the model; it is the model correctly reporting that 174 pallets cannot be delivered within the given time window with the current fleet.

4. Variables and Notation

The table below lists all symbols used in the equations that follow.

Symbol	What It Represents
V	All 26 nodes: V = {0, 1, …, 25}

Symbol	What It Represents
S	Store nodes only: S = {1, 2, …, 25}
K	The full fleet: K = K_heavy ∪ K_medium
Q^k	Carrying capacity of vehicle k (33p heavy, 12p medium)
q_i	Number of pallets that store i needs delivered
c_{ij}	Road distance between nodes i and j (km)
t_{ij}	Baseline free-flow travel time between i and j (minutes)
s_i	Time spent at store i = 10 + 2 × q_i (minutes)
$[e_i, l_i]$	Delivery window at store i: [480, 720] minutes past midnight
T_max	Maximum route duration: 405 minutes
μ(t)	Congestion multiplier applied to travel time at time t
x_{ij}^k	1 if vehicle k drives from node i to node j, 0 otherwise
T_i^k	Time vehicle k arrives at node i (minutes past midnight)
W_i^k	Waiting time of vehicle k at node i before the store opens
ε^k	Maximum allowed waiting time per scenario (45 / 75 / 90 min)
M	A very large number (99,999) used in the Big-M method

The two decision variables are x_ijk and T_ik — the unknowns the solver must find. x_ijk is binary (either 0 or 1); T_ik is continuous and may take any non-negative value. All other symbols are known parameters set before running the solver.

5. Objective Function

The goal of the model is to minimise the total kilometres driven by all trucks across all routes.

$$\text{Minimise} \quad Z = \Sigma_{k}\in K\ \Sigma_{i}\in V\ \Sigma_{j}\in V\ c_{ij} \cdot x_{ij}^{k}$$

This formula sums the distances of all arcs actually used. When x_ijk = 1, the distance c_ij for that arc is counted; when x_ijk = 0, the arc is unused and contributes nothing. Minimising Z drives the solver toward the shortest routes that still satisfy all constraints below.

Distance, rather than cost, was chosen as the objective because it is the most direct measure of routing efficiency and avoids the need to estimate or assume fuel prices, which can vary. Travel-time considerations are handled through the constraints rather than the objective.

6. Constraints

The constraints are the rules the solver must respect when finding routes. There are nine in total, ensuring the routes are physically possible, operationally valid, and legally compliant.

C1 — Every store must be visited exactly once

$$\Sigma_{k\in K} \Sigma_{j\in V} x_{ij}^{k} = 1 \quad \text{for all } i \in \{1, \ldots, 25\}$$

This is one of the most fundamental rules: each Lidl store receives exactly one delivery, from exactly one truck. Using ≥ 1 instead of = 1 would allow the solver to send two trucks to the same store, which does not reflect real deliveries. Setting it to exactly 1 avoids this.

C2 — What goes in must come out

$$\Sigma_{j\in V} x_{ij}^{k} - \Sigma_{j\in V} x_{ji}^{k} = 0 \quad \text{for all } i \in V, \text{ for all } k \in K$$

If a truck enters a node, it must also leave it. This applies to every node, including the depot. This constraint keeps routes connected and prevents the solver from building paths that start or end inside a store.

C3 — Each truck leaves the depot at most once

$$\Sigma_{j\in S} x_{0j}^{k} \leq 1 \quad \text{for all } k \in K$$

A truck can only depart from the depot once per day. Trucks that are not needed remain at the depot and are never activated. The ≤ 1 formulation allows for this possibility, unlike = 1, which would force every truck into service.

C4 — Each truck returns to the depot at most once

$$\Sigma_{i\in S} x_{i0}^{k} \leq 1 \quad \text{for all } k \in K$$

After completing its deliveries, each truck returns to the depot exactly once. Combined with C3, this ensures every active truck makes a single round trip from and back to node 0.

C5 — Trucks cannot carry more than they can hold

$$\Sigma_{i\in S}\, q_i \left(\Sigma_{j\in V} x_{ij}^{k}\right) \leq Q^{k} \quad \text{for all } k \in K$$

This sums all pallets assigned to a truck's route and checks that the total does not exceed what that truck can carry — 33 pallets for heavy trucks, 12 for medium trucks. Because each truck has its own Q_k value, this constraint automatically accommodates the different vehicle sizes without additional logic.

C6 — Arrival times must flow forward correctly with traffic

$$T_i^k + s_i + [\mu(T_i^k + s_i) \times t_{ij}] - M(1 - x_{ij}^k) \leq T_j^k$$

This is the most technical constraint in the model, and it does two things at once. First, it ensures arrival times move forward correctly along the route: when a truck leaves node i, its arrival at node j must account for the time already spent at i (arrival time T_ik), the service time s_i, and the traffic-adjusted travel time from i to j using μ(t). Second, it eliminates subtours — a subtour being a loop disconnected from the depot, such as a small circular route between three stores that never visits the depot. Because arrival times can only increase along the route, and the depot is the only starting point, any loop that does not connect back to the depot would require arrival times to move backwards at some point, which this constraint prevents.

The M term implements the Big-M technique, with M set to a very large number (99,999). When an arc is unused (x_ijk = 0), the term −M(1 − 0) = −M makes the left-hand side small enough that the constraint is automatically satisfied regardless of arrival times. When the arc is used (x_ijk = 1), the M term disappears and the constraint becomes active.

C7 — Stores must be visited during their opening hours

$$480 \leq T_i^k \leq 720 \quad \text{for all } i \in \{1, \ldots, 25\}, \text{ for all } k \in K$$

Every truck must arrive at a store between 08:00 (minute 480) and 12:00 (minute 720). If a truck arrives before 08:00 it waits, which is permitted; if it cannot reach a store before 12:00, that route is infeasible and the solver must find another combination.

C8 — Waiting time at stores is bounded

$$0 \leq W_i^k \leq \varepsilon^k \quad \text{for all } i \in \{1, \ldots, 25\}, \text{ for all } k \in K$$

When a truck arrives at a store before 08:00, it must wait; W_ik denotes the waiting time of truck k at store i. A maximum waiting limit ε_k is set per scenario — 45 minutes for S-99, 75 minutes for S-152, and 90 minutes for S-170. Without this constraint, the solver would reject otherwise valid routes simply because trucks arrive early; the upper limit likewise prevents the solver from accepting solutions where trucks sit idle for an unreasonable length of time.

This constraint was not present in the original formulation. It has been added here to fully document the model.

C9 — Variable types

$$x_{ij}^k \in \{0, 1\} \quad \text{and} \quad T_i^k \geq 0$$

x_ijk must be a binary integer — a truck either uses an arc or it does not, with no in-between. T_ik is a continuous variable that may take any non-negative real value. Because of the binary variables, this problem is classified as a Mixed-Integer Linear Programme (MILP), which is why a specialised solver such as OR-Tools is required rather than a simple linear programming solver.

7. Model Summary

The complete model is summarised below for reference.

$$\text{Minimise} \quad Z = \Sigma_k \in K \ \Sigma_i \in V \ \Sigma_j \in V \ c_{ij} \cdot x_{ij}^k \quad \text{subject to:}$$

Ref	Constraint	What It Enforces
C1	$\Sigma_{k,j}\, x_{ijk} = 1 \ \ \forall i \in S$	Each store visited by exactly one truck
C2	$\Sigma_j\, x_{ijk} - \Sigma_j\, x_{jik} = 0 \ \ \forall i,k$	Every truck that enters a node must leave it
C3	$\Sigma_j\, x_{0jk} \leq 1 \ \ \forall k$	Each truck departs the depot at most once
C4	$\Sigma_i\, x_{i0k} \leq 1 \ \ \forall k$	Each truck returns to the depot at most once
C5	$\Sigma_i\, q_i(\Sigma_j\, x_{ijk}) \leq Q_k \ \ \forall k$	Load on each truck stays within its capacity
C6	$T_{ik}+s_i+\mu(T_{ik}+s_i)\cdot t_{ij}-M(1-x_{ijk}) \leq T_{jk}$	Arrival times advance correctly with traffic
C7	$480 \leq T_{ik} \leq 720 \ \ \forall i \in S$	Stores visited within 08:00–12:00
C8*	$0 \leq W_{ik} \leq \varepsilon_k \ \ \forall i \in S$	Early arrivals can wait up to the slack limit
C9	$x_{ijk} \in \{0,1\},\ T_{ik} \geq 0$	Variable domains

** C8 is an addition to the original formulation.*

Network: 26 nodes — 1 depot (Node 0) + 25 Lidl stores (Nodes 1–25)
Fleet: 4 heavy trucks (33p each) + 4 medium trucks (12p each) = 180p total capacity
Decision variables: $x_{ijk} \in \{0,1\}$ for routing, $T_{ik} \geq 0$ for arrival times

Conclusion

This document describes the full mathematical model used for the Dr. Oetker delivery problem. The model comprises nine constraints covering every operational requirement: which stores are visited, how trucks move through the network, how much they can carry, when they arrive, how long they can drive, and how traffic affects their journey.

Compared to the original formulation, two additions were made. The first is the three-scenario framework (S-99, S-152, S-170), which was already being tested in the code but not documented in the equations. The second is Constraint C8, covering the waiting-slack parameter that was already present in the solver but had not been formally written down. Everything else in the original formulation was correct and has been retained exactly as it was.

The model is implemented in Python using Google OR-Tools with Guided Local Search, as described in the project solver file DrOetker_Lidl_Optimization_REALISTIC.py. The results for all three scenarios are documented in the Results folder of the repository.